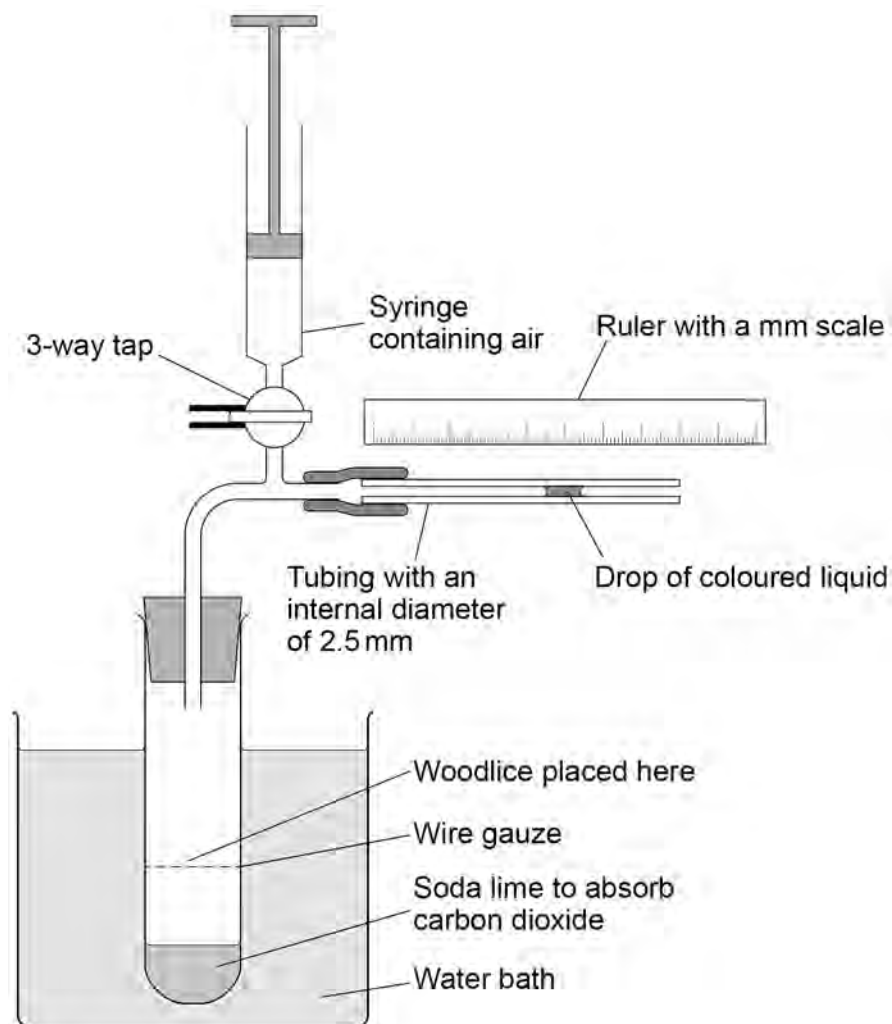


0 2

A student investigated the effect of temperature on the rate of aerobic respiration in woodlice.

Figure 1 shows the apparatus they used.

Figure 1



The student altered the temperature of the water bath and measured how far the drop of coloured liquid moved every minute for 5 minutes.

0 2

. 1

Other than those stated, suggest **two** variables the student should have kept constant in this investigation.

[1 mark]

1 _____

2 _____

0	2	.	2
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Describe how the student used the apparatus in **Figure 1** to reset the drop of coloured liquid back to the **right-hand** end of the tubing.

[2 marks]

0	2	.	3
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The student also set up a control experiment.

Suggest a suitable control experiment **and** explain why it was necessary.

[2 marks]

Control experiment _____

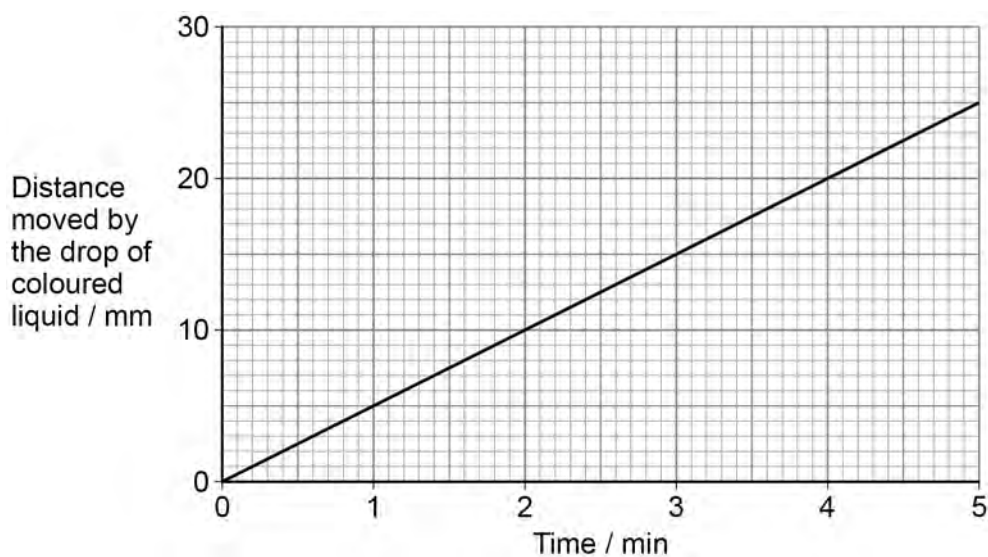
Explanation _____

Question 2 continues on the next page

0 2 . 4

Figure 2 shows the student's results at 25 °C

The mass of the woodlice in the tube was 3.78 g

Figure 2

Use **Figure 1** (on page 4) and **Figure 2** to calculate the mean rate of oxygen uptake by the woodlice in $\text{mm}^3 \text{s}^{-1} \text{g}^{-1}$

The formula for the volume of the capillary tubing is $\pi r^2 l$.

Use $\pi = 3.14$ in your calculation.

Show your working **and** give your answer to 2 decimal places.

[3 marks]Answer _____ $\text{mm}^3 \text{s}^{-1} \text{g}^{-1}$ 